INDIAN INSTITUTE OF INFORMATION TECHNOLOGY
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
COURSE: IMAGE AND VIDEO PROCESSING (IVP)

DeepGuard: An Explainable Deepfake Video Detection and Forensic Analysis Platform

Software Requirements Specification (SRS) & System Design Document
[Initial Project Proposal & Planning Submission]

Project Group: IVP_Group-10_DeepGuard

Team Members:
IIT2024108 | IIT2024109 | IIT2024110
IIT2024115 | IIT2024119 | IIT2024122

Document Release: Version 1.0 (Proposal & Planning Phase)
Primary Reference Paper: Lal, Shiwani & Gandhi (2025), Scientific Reports (Nature)
Date of Submission: October 2026

Document Control & Project Planning Metadata

Document Version	Project Phase	Primary Authors	Scope Summary
1.0 (Current)	Project Proposal & Planning	IVP Group 10 Team	Initial project formulation, requirements specification, proposed system architecture, and evaluation plan.

1. Introduction, Purpose and Scope

DeepGuard is a proposed visual deepfake video detection and digital forensic analysis platform designed to authenticate digital video media. With recent advancements in deep generative modeling, facial manipulation tools can seamlessly replace identities (Deepfakes, FaceSwap), modify facial expressions (Face2Face), and synthesize photorealistic mouth dynamics (NeuralTextures). These synthetic videos present severe societal risks, including political disinformation, executive impersonation, financial extortion, and reputation defamation. The purpose of this document is to establish the formal requirements, system architecture, data governance protocols, and testing methodology that will govern the development of the DeepGuard platform.

1.1 Scope of the Proposed System

The DeepGuard platform is proposed to encompass: (1) a deterministic temporal video ingestion and facial extraction pipeline utilizing MTCNN face detection; (2) an initial baseline neural network reproducing the visual soft-attention and BiLSTM architecture of Lal et al. (2025); (3) planned multi-branch architectural extensions incorporating mid-spatial boundary representations, high-frequency residual filters, and temporal Transformer modeling; (4) post-hoc statistical calibration via temperature scaling and out-of-distribution (OOD) distance estimation; (5) a high-performance RESTful API service; and (6) an interactive web dashboard for real-time video upload and forensic evidence inspection. The system will be developed under strict commodity hardware constraints, designed to operate entirely within 6.0 GB of VRAM on an NVIDIA GeForce RTX 3050 Laptop GPU.

1.2 Academic Reference Baseline

The primary methodological reference selected for this project is: K. Lal, S. Shiwani, and G. C. Gandhi, 'Deepfake video deception detection using visual attention-based method,' Scientific Reports, vol. 15, no. 1, art. 40089, 2025 (DOI: 10.1038/s41598-025-23920-0). The published paper proposes a 16-frame ResNeXt-50 spatial backbone paired with soft spatial attention and a 2048-hidden bidirectional LSTM, reporting published benchmark AUC scores of 0.962 on FaceForensics++ C23, 0.950 on Celeb-DF v2, and 0.920 on DFDC. DeepGuard proposes to reproduce this architecture as its foundational baseline, investigate its behavior under compression and unseen manipulations, and engineer robust architectural extensions to address identified generalization gaps.

2. Overall Product Perspective and Constraints

The proposed DeepGuard system is designed as a modular, client-server platform where video processing and neural network inference operate independently from model training.

2.1 Target Operating Environment & Hardware Constraints

To ensure academic reproducibility and practical deployment, all proposed models and pipelines will be engineered to function within commodity laboratory hardware: Target Hardware: NVIDIA GeForce RTX 3050 Laptop GPU (6.0 GB dedicated VRAM), 16 GB System RAM, Windows 11 / Linux x64. Software Stack: Python 3.11, PyTorch >= 2.0 with CUDA 12.x support, torchvision, OpenCV, facenet-pytorch, and FastAPI. Target Memory Envelope: Peak GPU VRAM allocation during video inference shall remain strictly below 1.0 GB (< 17% of total capacity). Target Latency: Model forward pass latency will target <= 35 ms per 16-frame sequence, with end-to-end video processing completing in under 3.5 seconds.

2.2 Proposed User Classes

• Forensic Media Analysts: Users requiring frame-level evidence, multi-branch confidence scores, and activation heatmaps to verify authentic vs. synthetic media.

• Academic Researchers & Students: Evaluators executing standardized benchmarks, running cross-dataset validation, and inspecting architectural ablations.

• General End-Users: Users seeking a clear, calibrated verdict (REAL or DEEPFAKE) through a straightforward drag-and-drop web dashboard.

3. Proposed Functional Requirements Specification

The following functional requirements specify the planned capabilities of the DeepGuard system. Each requirement is assigned a unique identifier, priority, and verification criterion.

Req ID	Proposed Operational Contract	Priority	Target Verification Method
FR-001	The system shall implement deterministic temporal frame sampling, extracting N=16 frames via uniform temporal stride or fixed random seed.	P0	Unit tests verifying deterministic index selection under fixed seed.
FR-002	The system shall integrate MTCNN face detection to localize facial regions with a planned 0.20 (20%) bounding box margin expansion and bilinear resizing to 112x112.	P0	Automated tests asserting output tensor shape (B, 16, 3, 112, 112).
FR-003	The system shall implement the foundational reference baseline: ResNeXt-50 (32x4d) + Soft Spatial Attention + 2-layer Bidirectional LSTM (2048 hidden units).	P0	Layer-by-layer parameter count and tensor dimension contract tests.
FR-004	The system shall design a multi-branch feature decomposition extension incorporating mid-spatial boundary features, high-frequency residual filters, and	P0	Gradient flow tests verifying all parallel branches backpropagate gradients.

	temporal Transformer modeling.		
FR-005	The system shall provide a modular model registry supporting independent swapping between the baseline model and research extensions.	P0	Predictor factory tests confirming dynamic model instantiation via registry keys.
FR-006	The system shall implement post-hoc temperature scaling to calibrate raw logits into statistically meaningful probabilities.	P0	Validation script demonstrating reduction of Expected Calibration Error (ECE <= 10%).
FR-007	The system shall calculate penultimate feature Mahalanobis / Euclidean distance against a validation manifold threshold to flag out-of-distribution (OOD) inputs.	P0	Unit tests asserting OOD flag is raised on perturbed/novel distribution inputs.
FR-008	The system shall establish two operational decision policies: standard balanced classification (tau = 0.50) and a high-sensitivity policy (tau* guaranteeing >=90% clean specificity).	P0	Threshold policy validation asserting >=90% clean specificity on authentic validation set.
FR-009	The system shall provide a FastAPI REST service exposing endpoints for single-video prediction (/api/v1/predict), model metadata (/api/v1/model-info), and service health (/health).	P0	API test client suite asserting HTTP 200 responses and JSON schema compliance.
FR-010	The system shall develop an interactive dark-mode web dashboard supporting drag-and-drop video upload, real-time threshold adjustment, and latency breakdown display.	P0	Browser interface validation asserting DOM rendering and asynchronous fetch completion.
FR-011	The system shall provide a command-line interface (CLI) supporting prediction, evaluation across manifests, and model serving.	P0	CLI invocation tests asserting exit code 0 across subcommands.
FR-012	The system shall support exporting trained model weights to ONNX and TorchScript runtime formats with high numerical parity.	P1	Parity tests asserting maximum absolute difference between PyTorch and ONNX < 1e-5.
FR-013	The system shall enforce an indivisible group manifest splitting utility ensuring all video sequences derived from the same source subject reside strictly in the same split.	P0	Manifest integrity test asserting zero group overlap between train and test partitions.
FR-014	The system shall implement a leakage-safe 5-fold cross-validation partitioner to satisfy academic evaluation requirements.	P1	Group-K-Fold unit tests asserting zero subject overlap across all 5 generated folds.
FR-015	The system shall incorporate explainable attribution mechanisms producing spatial attention weight heatmaps	P1	Attribution generator asserting heatmap array shape matches input spatial

	across facial frames.		dimensions.
FR-016	The system shall implement late-stage log-odds evidence fusion to combine predictions from complementary spatial and spectral detector streams.	P1	Fusion contract tests asserting monotonicity and convexity constraints.

4. Proposed Non-Functional Requirements Specification

Req ID	Category	Proposed Performance Target	Planned Verification Strategy
NFR-001	Latency	Model forward pass P50 latency shall be <= 35.0 ms per 16-frame sequence on NVIDIA RTX 3050 GPU.	Benchmark script executing 100 consecutive forward passes with CUDA synchronization.
NFR-002	Throughput	End-to-end video processing latency shall complete within <= 3.5 seconds per 16-frame video clip.	Wall-clock timing covering OpenCV decoding, MTCNN face extraction, forward pass, and reporting.
NFR-003	Memory	Peak dedicated GPU memory allocation shall not exceed 1.0 GB during video inference.	PyTorch torch.cuda.max_memory_allocated() profiling during inference.
NFR-004	Determinism	Given identical random seeds, repeated inference on the same input video shall produce bitwise identical logits.	Automated test asserting delta logits == 0.000000 across multiple runs.
NFR-005	Numerical Parity	Exported ONNX runtime models shall match PyTorch forward pass outputs within absolute error < 1e-5.	Numerical difference validation comparing ONNX Runtime outputs with PyTorch tensors.
NFR-006	Data Integrity	Zero data leakage shall occur between training, validation, and test partitions across all datasets.	Group-level validation scripts verifying disjoint sets of source identities.
NFR-007	Calibration	The model shall achieve an Expected Calibration Error (ECE) <= 10.0% on held-out validation data post-calibration.	Reliability curve computation binning predictions into 10 probability intervals.
NFR-008	Clean Specificity	Under the high-sensitivity threshold policy, the detector shall achieve >= 90.0% specificity on clean authentic media.	Evaluation on authentic video holdouts asserting False Positive Rate <= 10.0%.

5. Proposed System Architecture and Module Design

The proposed DeepGuard architecture decomposes the video forensics lifecycle into modular, decoupled subsystems:

5.1 Video Ingestion & Deterministic Sampling Module

The video ingestion module will accept video containers in MP4, AVI, MOV, or MKV formats. Using OpenCV / Decord, the module will inspect container metadata and verify that the video contains at least 16 decodable frames. A deterministic uniform temporal sampler will extract N=16 frames distributed evenly across the video duration. To ensure reproducibility during testing, random sampling will support seed derivation based on the SHA-256 hash of the video file.

5.2 MTCNN Face Localization & Preprocessing Module

Extracted raw frames will be processed by a Multi-Task Cascaded Convolutional Network (MTCNN) to detect facial bounding boxes and 5 facial landmarks. To ensure that subtle boundary blending artifacts along the jawline, forehead, and ears are captured, the bounding box will be expanded by 20% (margin factor = 0.20). The isolated facial region will be bilinearly resized to 112 x 112 x 3 pixels and standardized using ImageNet mean [0.485, 0.456, 0.406] and standard deviation [0.229, 0.224, 0.225]. The output will be formatted as a 5D sequence tensor of shape (B, 16, 3, 112, 112). If no face is detected, a fallback center crop will be applied to prevent pipeline crashing.

5.3 Baseline Feature Extraction & Attention Module

Following the reference paper (Lal et al., 2025), the foundational model will employ an ImageNet-pretrained ResNeXt-50 (32x4d) backbone. The spatial convolutional stages will output feature maps of shape (B*T, 2048, 4, 4). A soft spatial attention module using a 1x1 convolution will compute spatial importance weights across the 4x4 grid via spatial softmax. A weighted sum across spatial coordinates will yield attended 2048-dimensional feature vectors per frame.

5.4 Multi-Branch Research Extension

To overcome the known limitation where boundary detectors fail on expression reenactment (Face2Face), DeepGuard proposes a multi-branch feature decomposition: (1) A mid-spatial branch extracting Stage 5 features from an EfficientNet-B0 backbone to preserve localized boundary seams; (2) A high-frequency residual branch applying Laplacian and Spatial Rich Model (SRM) filter banks to isolate compression-resistant frequency discrepancies; and (3) A texture anomaly branch analyzing spatial gradient variance. These multi-domain representations will be aggregated across time using a 4-head Temporal Transformer encoder.

5.5 Statistical Calibration & Decision Layer

To prevent misleading overconfident predictions, raw logits z will be calibrated using post-hoc temperature scaling: p_cal = sigmoid(z / T). The scalar temperature T will be fitted strictly on held-out validation sequences by minimizing negative log-likelihood (NLL). Additionally, an out-of-distribution (OOD) gating estimator will compute the regularized Mahalanobis distance of penultimate features against an empirical validation threshold (3.18 at 99th percentile), flagging anomalous or out-of-domain media.

5.6 FastAPI Service & Web Forensics Dashboard

The backend will be deployed as an asynchronous FastAPI microservice exposing multipart upload endpoints. The user interface will be developed as a lightweight, dependency-free dark-mode web dashboard featuring: a video player preview, real-time threshold slider (toggling between standard tau=0.50 and high-sensitivity tau*), side-by-side model comparison, and a processing latency waterfall breakdown.

6. Dataset Requirements & Anti-Leakage Protocol

The DeepGuard evaluation plan requires three distinct benchmark datasets:

• FaceForensics++ (FF++) C23: Primary benchmark dataset consisting of 1,000 real YouTube videos and 4,000 manipulated videos across Deepfakes, Face2Face, FaceSwap, and NeuralTextures, compressed at H.264 CRF 23. This dataset will serve for in-domain training, validation, and Leave-One-Family-Out cross-manipulation matrix benchmarking.

• Celeb-DF v2: External evaluation dataset containing 518 official test videos (178 real, 340 synthetic) exhibiting advanced boundary blending. Celeb-DF v2 will be reserved strictly for zero-shot cross-dataset generalization testing without retraining.

• Deepfake Detection Challenge (DFDC): Planned external evaluation benchmark comprising ~100,000 videos. Access will depend on Kaggle competition credential authorization.

• Indivisible Group Partitioning Invariant: All dataset splits will be partitioned strictly by donor subject identity (group_id). Under no circumstances will manipulated derivatives of a subject appear in a different partition from the pristine source video, guaranteeing zero identity and blending leakage.

7. Proposed Database & API Design

The system will define lightweight relational entities to track forensic analyses: videos (id, filename, sha256_hash, duration, fps), analyses (id, video_id, engine_key, threshold, calibrated_score, verdict, ood_flag, created_at), and frame_evidence (analysis_id, frame_no, timestamp, attention_score).

Proposed REST Endpoints:

• POST /api/v1/predict — Multipart video file upload. Returns predicted verdict, calibrated probability, OOD flag, and latency metrics.

• POST /api/v1/compare — Dual-engine execution comparing the baseline ResNeXt model against the proposed multi-branch model.

• GET /api/v1/model-info — Returns active model engine metadata, parameters, and limitation disclosures.

• GET /health — Returns GPU allocation, active device, and memory consumption.

8. Technical Risks, Feasibility & Mitigation Strategy

Identified Risk	Impact	Planned Technical Mitigation
GPU VRAM Exhaustion on 6GB	High	Adopt mixed precision (FP16/AMP), micro-batching (batch size 4), gradient

RTX 3050		accumulation (2 steps), and frozen backbone stems.
Identity / Subject Data Leakage	Critical	Enforce indivisible group_id partitioning; verify disjoint subject sets via automated unit tests prior to training.
Generalization Drop on Unseen Forgeries	High	Implement Leave-One-Family-Out (LOFO) matrix benchmarking and multi-branch frequency/texture representations.
Overconfident Probabilities / False Alarms	High	Incorporate post-hoc temperature scaling fitted strictly on validation data, and derive operating thresholds guaranteeing >=90% clean specificity.
Face Detection Failure on Occluded Faces	Medium	Implement fallback central cropping and landmark quality verification in the preprocessing module.
External Dataset Access Restrictions	Medium	Develop modular dataset adapters (e.g. for DFDC) and maintain standalone manifests allowing offline evaluation once data is authorized.

9. Planned Activity Timeline & 16-Week Phase Plan

Project Phase	Scheduled Duration	Key Planned Activities & Milestone Objectives
Phase 1: Requirements & Literature	Weeks 1 – 2	Formalize SRS; conduct comprehensive survey of 18 peer-reviewed papers; analyze base paper methodology.
Phase 2: Dataset Acquisition & Pipeline	Weeks 3 – 4	Obtain FaceForensics++ C23 access; implement OpenCV decoding, uniform sampling, MTCNN face localization, and group splitting.
Phase 3: Baseline Model Reproduction	Weeks 5 – 7	Implement ResNeXt-50 + Soft Attention + BiLSTM; train on FF++ C23 under mixed precision; evaluate baseline metrics.
Phase 4: Multi-Branch Invariant Design	Weeks 8 – 10	Develop mid-spatial Stage 5, high-frequency residual, and texture branches; integrate Temporal Transformer; execute 4-fold LOFO matrix.
Phase 5: Calibration & Forensics	Weeks 11 – 12	Fit post-hoc temperature scaling; implement penultimate Mahalanobis OOD gating; evaluate cross-dataset transfer on Celeb-DF v2.
Phase 6: Serving & Dashboard	Weeks 13 – 14	Build FastAPI microservice; develop interactive dark-mode web dashboard; implement ONNX / TorchScript model export.
Phase 7: Testing & Documentation	Weeks 15 – 16	Execute comprehensive regression test suite; finalize design reports, presentation deck, and academic submission package.